PROJECT PLANNING

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

Rising Waters is an AI-powered Flood Prediction System that predicts flood risk using rainfall and weather parameters. The project aims to provide an accurate, reliable, and user-friendly solution for early flood prediction and disaster preparedness.

Project Objectives

- Predict flood risk accurately using Machine Learning.
- Analyze rainfall and environmental conditions.
- Develop a user-friendly Streamlit web application.
- Support government authorities with early warning systems.

Project Scope

The project focuses on collecting rainfall and weather data, processing the information using Machine Learning algorithms, and displaying flood risk predictions through a web dashboard. The application can be used by disaster management teams and weather departments.

Project Timeline

Week 1
Requirement Analysis and Dataset Collection

Week 2
Data Cleaning and Preprocessing

Week 3
Feature Engineering and Model Training

Week 4
Model Evaluation and Accuracy Improvement

Week 5
Streamlit Application Development

Week 6
Testing and Bug Fixing

Week 7
Deployment using Streamlit Cloud

Week 8
Documentation and Final Presentation

Project Resources

Software
• Python
• Streamlit
• Pandas
• NumPy
• Scikit-learn
• XGBoost
• Joblib
• GitHub

Hardware
• Intel Core i3 or above
• 4 GB RAM
• 10 GB Storage
• Internet Connection

Project Deliverables

• Trained Machine Learning Model
• Rising Waters Web Application
• GitHub Repository
• Streamlit Deployment
• Project Documentation
• Final PPT Presentation

Risk Management

• Missing dataset values
• Incorrect user input
• Low model accuracy
• Deployment issues

These risks are minimized through data preprocessing, model validation, and extensive application testing.

Expected Outcome

The Rising Waters application will provide fast and accurate flood predictions, helping authorities make timely decisions and reduce the impact of floods.

Conclusion

A proper project plan ensures systematic development, timely completion, and successful deployment of the Rising Waters application. The planning phase acts as the roadmap for the entire project.